Vedant Rupwal

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Professional Summary

Math and Computer Science student with a strong foundation in computational modeling, machine learning, and spatiotemporal data analysis. Experienced in migrating complex mathematical algorithms to Python and developing geospatial features for consumer-facing applications. Seeking to leverage expertise in Data Science and AI to deliver high-impact, data-driven solutions.

Education

University of Illinois at Chicago | Chicago, IL

- Bachelor of Science in Math and Computer Science | 2023 - 2026
- Minor in Economics | 2023 - 2026

Experience

AI & Data Advisor | Eigen Axis

Dec 2025 - Present

- Systems Architecture: Architected a production-ready, full-stack web application and Retrieval-Augmented Generation (RAG) backend tailored for clinical knowledge discovery and managing LLM training pipelines.
- Data Engineering: Engineered an automated, multi-channel data ingestion pipeline with exponential jittered backoff and a Dead Letter Queue (DLQ), reliably extracting structured texts from Wikipedia APIs, PubMed, and local PDFs.
- Search Infrastructure: Developed a hybrid embedding engine combining dense neural vectors (Sentence-Transformers) and statistical BM25 sparse matrices, streaming real-time indexing into a localized Qdrant vector database.
- Model Deployment: Integrated PyTorch and TensorFlow training loops with a FastAPI backend and WebSockets, enabling real-time monitoring of model states (Idle, Training, Running) and seamless multi-environment deployments.
- Security & Compliance: Built a highly secure authentication domain with TOTP MFA, PBKDF2-HMAC-SHA256 password hashing, and Role-Based Access Control (RBAC) to govern schema-driven YAML configurations and track system audit logs.

Data Science Research Assistant | University of Illinois, Chicago

Jan 2025 – May 2026

- Algorithm Migration: Pioneering the migration of complex MATLAB research codebases to Python, improving the scalability and efficiency of data processing pipelines, and improving processing time by 98% (from 3 months to 30 hours).
- Spatiotemporal Analysis: Performing advanced spatiotemporal mapping of topography to analyze the thickness of transition in ultrathin foam films.
- Statistical Modeling: Analyzing raw data to derive meaningful insights and visualize physical properties using mathematical modeling.

Experience (Cont.)

AI Developer (Spark Hacks) | Chicago, IL
2025

- Geospatial Engineering: Engineered a real-time mapping feature that connected users with food vendors within their locality, anticipating a 20% increase in vendor sales through proximity discovery.
- Sustainable Logistics: Designed backend logic for handling dynamic inventory data, reducing food waste through real-time transactions between surplus and demand.

Personal Project

Ask the Pandit | Full-Stack RAG DeveloperTech Stack: Python, JavaScript, HTML/CSS, FastAPI, ChromaDB, Hugging Face, OpenAI SDK, Qwen 2.5

- End-to-End Application: Developed a context-aware Chrome extension and highly optimized FastAPI backend to deliver precise, text-backed RAG answers sourced from extensive theological scriptures.
- Database Partitioning: Architected a dynamic hybrid database routing system using ChromaDB to aggressively split massive datasets, optimizing system memory limits and accelerating semantic search speed.
- Advanced Retrieval: Implemented a two-stage retrieval pipeline utilizing all-MiniLM-L6-v2 for rapid similarity search and ms-marco-MiniLM-L-6-v2 CrossEncoders to accurately rerank top vectors for deep reading comprehension.
- Dynamic UI Integrations: Built a Shadow DOM-injected interface that intelligently scrapes active webpage elements, feeding up to 6,000 characters of real-time contextual reading material into the LLM.
- Safety Engineering: Designed strict, prompt engineering guardrails, including zero hallucination locks and false-premise handlers, to restrict AI responses to verified database texts and correct logical traps.

Certifications

- Accelerated Deep Learning with GPU – Jun 2025
- Deep Learning using TensorFlow (Gen AI focus) – Jun 2025
- Applied Data Science with R (Level 2) – Jun 2025
- Data Visualization with R – Jun 2025
- Using R with Databases – Jun 2025
- Agentforce Innovator - Jun 2025

Technicall Skills

- Artificial Intelligence & ML: Machine Learning Models, Predictive Modeling, Data Classification
- Data Science: Spatiotemporal Mapping, Statistical Analysis, Data Visualization, Python, R, MATLAB
- Languages: Python, C/C++, Java, R, MATLAB
- Core Competencies: Data Structures & Algorithms (DSA), Code Migration, Leadership